Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following orders of acid strength is correct?

(A) $CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH < CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH < CH_2\underset{\underset{Cl}{|}}{CH}CH_2COOH$

(B) $CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH > CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH > CH_2\underset{\underset{Cl}{|}}{CH}CH_2COOH$

(C) $CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH > CH_2\underset{\underset{Cl}{|}}{CH}CH_2COOH > CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH$

(D) $CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH < CH_2\underset{\underset{Cl}{|}}{CH}CH_2COOH < CH_3\underset{\underset{Cl}{|}}{CH}CH_2COOH$

Q.2

Which of the following statements is not correct?

(A) Carboxylic acids and their derivatives are collectively known as acyl compounds.

(B) Acyl compounds (RCOW) undergo electrophilic substitution reaction in which W is replaced by some other basic group.

(C) Acyl compounds undergo nucleophilic substitution reactions more readily than the compounds containing no carbonyl group.

(D) The acyl compounds differ from aldehydes or ketones in the respect that the former undergo nucleophilic substitution reactions while the latter undergo nucleophilic addition reactions.

Q.3

Acetamide on heating with phosphorus pentoxide gives

(A) CH_3COOH

(B) CH_3NH_2

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(C) CH_3CN

(D) $CH_3CH_2NH_2$

Q.4

Which of the following statements is correct?

(A) Formic acid like acetic acid can be halogenated in the presence of red phosphorus and Cl_2.

(B) Formic acid is a stronger acid than acetic acid.

(C) The boiling point of propanoic acid is less than that of n-butyl alcohol, an alcohol of comparable molar mass.

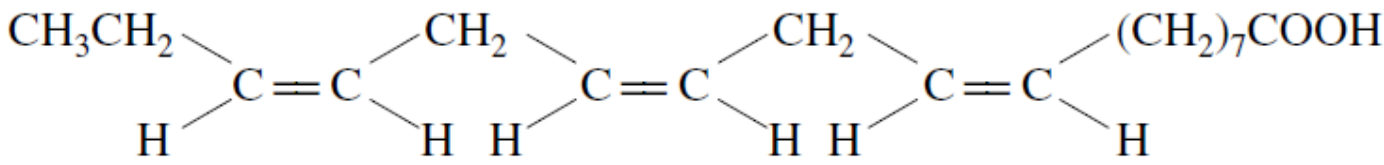
(D) The IUPAC name of $CH_3CH_2CH_2CN$ is propanenitrile.

Q.5

Which of the following statements is not correct?

(A) The melting point of a trans fatty acid is higher than its cis isomer.

(B) The IUPAC name of



is *cis*,*cis*,*cis*-3,6,9-octadecatrien-18-oic acid.

(C) Vinegar is a 5% aqueous solution of acetic acid.

(D) The $C = O$ bond in $RCOOH$ is longer than in $RCHO$.

Q.6

The treatment of an ester with $LiAlH_4$ followed by acid hydrolysis produces

(A) two alcohols

(B) two acids

(C) two aldehydes

(D) one molecule of alcohol and another of carboxylic acid

Q.7

The Hell-Volhard-Zelinsky reaction is used to synthesise

(A) aldehydes

(B) α-haloacids

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(C) acid halides

(D) nitriles

Q.8

Which of the following reagent solutions can be used to distinguish between methanoic acid and ethanoic acid?

(A) Ammoniacal $AgNO_3$ solution

(B) $FeCl_3$ solution

(C) Na_2CO_3 solution

(D) $NaOH$ solution

Q.9

The reaction of formic acid with concentrated H_2SO_4 gives

(A) CH_3COOH

(B) CO_2

(C) $HCHO$

(D) CO

Q.10

Which of the following sequences of relative rates of alkaline hydrolysis of esters is correct?

(A) $CH_3COOCH_3 > CH_3COOC_2H_5 > CH_3COOC_3H_7$

(B) $CH_3COOCH_3 < CH_3COOC_2H_5 < CH_3COOC_3H_7$

(C) $CH_3COOCH_3 < CH_3COOC_3H_5 > CH_3COOC_3H_7$

(D) $CH_3COOCH_3 > CH_3COOC_2H_5 < CH_3COOC_3H_7$

Q.11

Which of the following statements is not correct?

(A) Maleic acid is a weaker acid than fumaric acid.

(B) Maleate monoanion is a weaker acid than fumarate monoanion.

(C) Salicyclic acid, *o* - HOC_6H_4COOH, is a stronger acid than *o* $- H_3COC_6H_4COOH$.

(D) A more reactive acid derivative may be used to prepare a less reactive one by reaction with the appropriate nucleophile.

Q.12

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Which of the following reactants would produce acetophenone on heating?

(A) Calcium acetate + Calcium formate

(B) Calcium formate + Calcium benzoate

(C) Calcium acetate + Calcium benzoate

(D) Calcium acetate

Q.13

Which of the following reagent/solution can be used to distinguish between methanoic acid and ethanoic acid?

(A) Tollens reagent

(B) HCl solution

(C) NaOH solution

(D) $\mathrm{NaHCO_3}$ solution

Q.14

Which of the following orders is true regarding the boiling point of organic substances?

(A) $\mathrm{CH_3COCl} > \mathrm{CH_3COOH} > (\mathrm{CH_3CO})_2\,\mathrm{O}$

(B) $\mathrm{CH_3COCl} < \mathrm{CH_3COOH} < (\mathrm{CH_3CO})_2\,\mathrm{O}$

(C) $\mathrm{CH_3COCl} > \mathrm{CH_3COOH} < (\mathrm{CH_3CO})_2\,\mathrm{O}$

(D) $\mathrm{CH_3COCl} < \mathrm{CH_3COOH} > (\mathrm{CH_3CO})_2\,\mathrm{O}$

Q.15

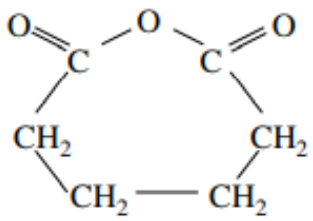
Which of the following statements is not correct?

(A) The ease with which :W of the acyl compound RCOW is lost depends upon its basicity; the weaker the base, the better the leaving group.

(B) Aldehydes or ketones undergo nucleophilic addition reactions instead of substitution reactions because the leaving group ($:\mathrm{H}^-$ or $:\mathrm{R}^-$) are the strongest bases of all.

(C) It is more convenient to prepare amides and esters by first converting carboxylic acid into acid chloride followed by its reaction with ammonia or an alcohol.

(D) Adipic acid $\mathrm{HOOC\,(CH_2)_4\,COOH}$ on heating produces cyclic adipic anhydride having structure.



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Q.16

Which of the following statements is correct?

(A) An electron-releasing group G in $p-\mathrm{GC_6H_4COOC_2H_5}$ decreases the rate of hydrolysis as compared to $\mathrm{C_6H_5COOC_2H_5}$

(B) The rate of alkaline hydrolysis of RCOOR' increases with the increase in the carbon atoms in the alkyl group $\mathrm{R'}$

(C) The rate of alkaline hydrolysis of $\bar{R}$ COOR' increases with the increase in the carbon atoms in the alkyl group $\bar{R}$.

(D) The rate of alkaline hydrolysis of RCOOR' decreases with the increase in the branching of carbon skeleton of the alkyl group R.

Q.17

The oxidation products of 1-nitronaphthalene and α-naphthylamine respectively are

(A) phthalic acid and 3-aminophthalic acid

(B) 3-nitrophthalic acid and phthalic acid

(C) phthalic acid and phthalic acid

(D) 3-nitrophthalic acid and 3 -aminophthalic acid

Q.18

Acetic acid differs from formic acid in that

(A) acetic acid is stable to heat

(B) formic acid is stable to heat

(C) acetic acid acts as a reducing agent

(D) acetic acid shows positive test with Tollens reagent

Q.19

Which of the following orders is true regarding the acetic nature of monosubstituted acetic acid?

(A) Fluoroacetic acid $>$ Chloroacetic acid $>$ Bromoacetic acid

(B) Fluoroacetic acid $<$ Chloroacetic acid $<$ Bromoacetic acid

(C) Fluoroacetic acid $<$ Chloroacetic acid $>$ Bromoacetic acid

(D) Fluoroacetic acid $<$ Chloroacetic acid $<$ Bromoacetic acid

Q.20

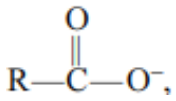
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Which of the following statements is not correct?

(A) The reduction of RCOOH with $LiAlH_4$ produces RCH_3.

(B) The - COOH group attached to benzene ring is meta director.

(C) In



both the carbon-oxygen bond lengths are identical. It is more than carbon-oxygen double bond length, and less than carbon-oxygen single bond length.

(D) Electron-withdrawing substituents stabilises the carboxylate anion and hence increase the acidity of carboxylic acid.

Q.21

When propionic acid is treated with aqueous sodium bicarbonate, CO_2 is liberated. The C from CO_2 comes from

(A) methyl group

(B) carboxylic acid group

(C) methylene group

(D) bicarbonate

Q.22

Which of the following statements is correct?

(A) Alkaline hydrolysis of an ester is a reversible reaction.

(B) In the alkaline hydrolysis of an ester, the bond broken is the bond between oxygen and alkyl group.

(C) Alkaline hydrolysis of an ester follows second order kinetics and is thus S_N2.

(D) An electron-withdrawing group G in $p-GC_6H_4COOC_2H_5$ decreases the rate of hydrolysis as compared to $C_6H_5COOC_2H_5$

Q.23

Which of the following statements is not correct?

(A) $H_2C=CHCH_2COOH$ is more acidic than CH_3CH_2COOH.

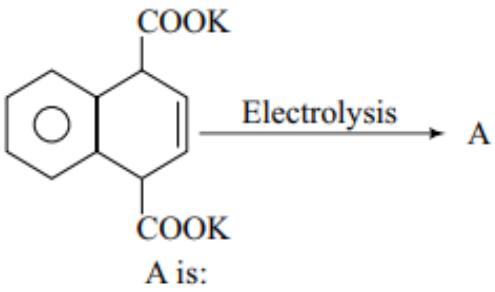
(B) The acid $(CH_3)_3CCH_2C(CH_3)_2COOH$ is less acidic than $CH_3(CH_2)_7COOH$.

(C) Trans-4-t-butylcyclohexanecarboxylic acid is stronger than the cis-isomer

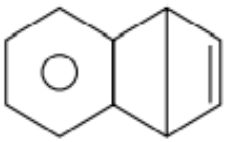
(D) The pK_{a1}° of a dicarboxylic acid is usually greater than the pK_a° of the comparable monoalkylcarboxylic acid.

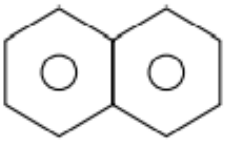
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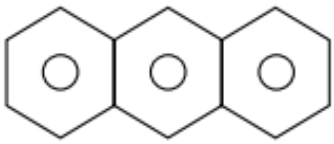
Q.24

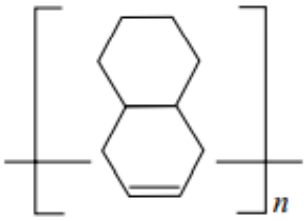


A is:

(A) 

(B) 

(C) 

(D) 

Q.25

The treatment of CH_3CH_2COOH with chlorine in the presence of phosphorus gives

(A) CH_3CH_2COCl

(B) $CH_3CH_2CH_2Cl$

(C) $CH_3CH(Cl)COOH$

(D) $CH_2(Cl)CH_2COOH$

Answers & Solutions

Q.1 Answer:

$$\underset{\displaystyle Cl}{CH_3CH_2CHCOOH} > \underset{\displaystyle Cl}{CH_3CHCH_2COOH} > \underset{\displaystyle Cl}{CH_2CH_2CH_2COOH}$$

Solution:

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Farther Cl atom from COOH group, weaker the acid.

Q.2 Answer:

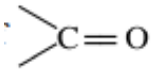
Acyl compounds (RCOW) undergo electrophilic substitution reaction in which W is replaced by some other basic group.

Solution:

Explaining the options

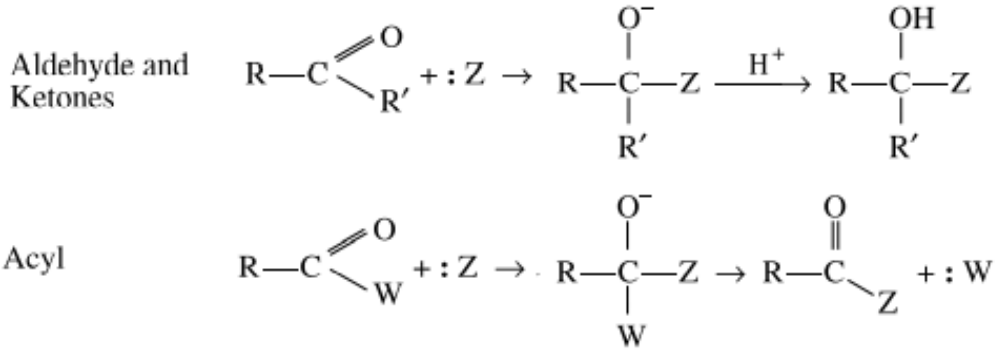
(a) The acyl compounds contain RCO group.

(b) Because of



group, acyl compounds undergo nucleophilic substitution reactions because carbonyl carbon is slightly positively charged.

(d)

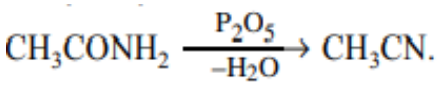


So, the correct option is (b) since the statement is wrong

Q.3 Answer:

CH_3CN

Solution:



Q.4 Answer:

Formic acid is a stronger acid than acetic acid.

Solution:

Explaining the options

(a) Formic acid does not contain α-hydrogen atom.

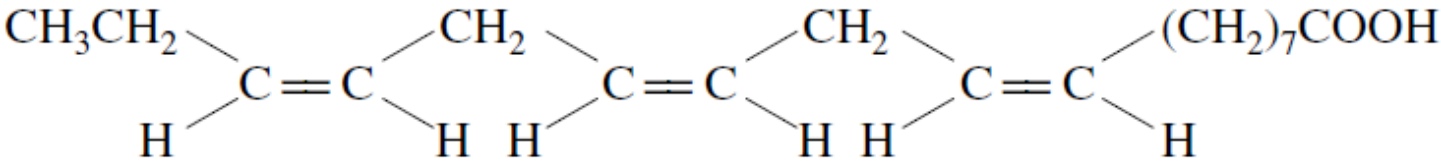
(c) The boiling point of carboxylic acid is higher than that of alcohol of comparable molar mass because of more extensive hydrogen bonding.

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(d) The - CN group is also counted in the carbon chain. Hence, its name is butanenitrile. So, the only correct option is (b)

Q.5 Answer:

The IUPAC name of



is *cis*,*cis*,*cis*-3,6,9-octadecatrien-18-oic acid.

Solution:

(b) Counting starts with COOH group. The correct name is cis, cis, cis $-9, 12, 15$-octadecatrienoic acid. The statements in the given options are correct.

Q.6 Answer:

two alcohols

Solution:

Two alcohols are produced. $\mathrm{RCOOR' \xrightarrow{LiAlH_4} RCH_2OH + R'OH.}$

Q.7 Answer:

α-haloacids

Solution:

The Hell-Volhard-Zelinsky reaction is used to synthesise α-haloacids.
HVZ reaction is used for the
synthesis of (b) α-bromo acids.
When carboxylic acids are made to react with bromine in the presence of phosphorus, it forms α-bromo acids.
HVZ reaction stands for Hell-Volhard-Zelinsky Reaction. It is for the purpose of halogenation of carboxylic acids at the α carbon.

Q.8 Answer:

Ammoniacal $\mathrm{AgNO_3}$ solution

Solution:

Methanoic acid (i.e. formic acid) reduces ammoniacal $\mathrm{AgNO_3}$ solution (Tollens reagent) while ethanoic acid does not.

Q.9 Answer:

CO

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Solution:

$$HCOOH \xrightarrow{-H_2O} CO$$

Q.10 Answer:

$CH_3COOCH_3 > CH_3COOC_2H_5 > CH_3COOC_3H_7$

Solution:

Increasing bulkiness of R' group in RCOOR' decreases the rate of alkaline hydrolysis of ester.

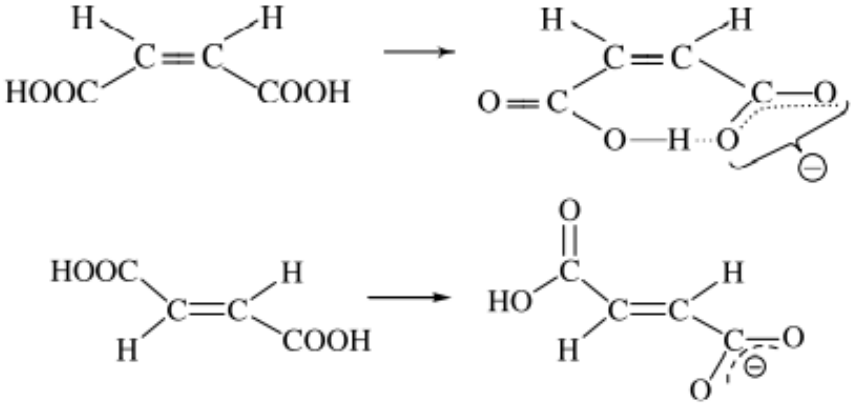
Q.11 Answer:

Maleic acid is a weaker acid than fumaric acid.

Solution:

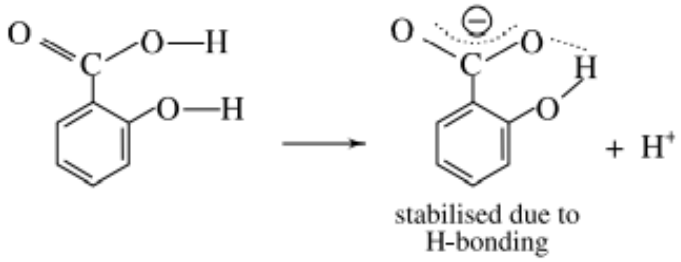
Expalining the option

(a) The maleic acid is a stronger acid because of the stabilisation of its monoanion due to H-bonding between the cis $COOH$ and COO^-. This is not present in the trans isomer (fumarate anion).



(b) H-bonding in maleate monoanion makes the removal of H more difficult as compared to fumarate monoanion where H-bonding is not present.

(c) H-bonding in salicylate ion makes the salicyclic acid more acidic than $O-$ $H_3COC_6H_4COOH$.



Q.12 Answer:

Calcium acetate + Calcium benzoate

Solution:

Acetophenone is produced on heating calcium acetate and calcium benzoate.

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Q.13 Answer:

Tollens reagent

Solution:

Methanoic acid (i.e. formic acid) reduces ammoniacal $AgNO_3$ solution (Tollens reagent) while ethanoic acid does not.

Q.14 Answer:

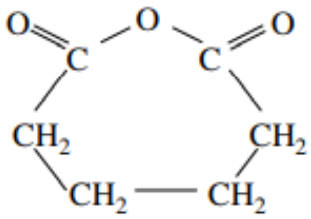
$CH_3COCl < CH_3COOH < (CH_3CO)_2O$

Solution:

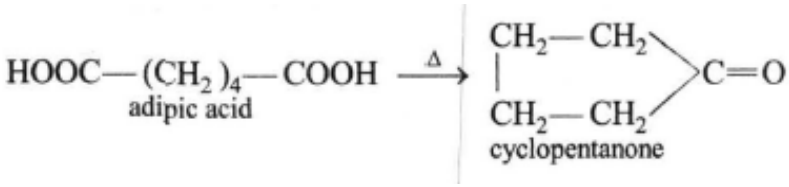
CH_3COCl does not involve hydrogen bonding. Its boiling point is lesser than CH_3COOH. The boiling point of the latter is lesser than that of $(CH_3CO)_2O$ due to larger molar mass of $(CH_3CO)_2O$.

Q.15 Answer:

Adipic acid $HOOC(CH_2)_4COOH$ on heating produces cyclic adipic anhydride having structure.



Solution:



So, the correct option is (d)

Q.16 Answer:

An electron-releasing group G in $p-GC_6H_4COOC_2H_5$ decreases the rate of hydrolysis as compared to $C_6H_5COOC_2H_5$

Solution:

(a) An electron-releasing group destabilises the intermediate carboxylate anion, hence decreases the rate of hydrolysis. So, this option is not correct. The remaining options are correct.

Q.17 Answer:

3-nitrophthalic acid and phthalic acid

Solution:

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Nitro bearing benzene ring is resistant to oxidation and amino bearing benzene is more susceptible to oxidation. The products are 3-nitrophthalic acid and phthalic acid, respectively.

Q.18 Answer:

acetic acid is stable to heat

Solution:

Acetic acid is thermally more stable than formic acid.

Formic acid is more acidic than acetic acid because the elimination of $\mathrm{H}+$ ion is not opposed by any group and obtained formate ion is resonance stabilized.

In Acetic acid, the elimination of $\mathrm{H}+$ ion is opposed by $+$ leffect of - CH3 group attached to $-\mathrm{COOH}$ group.

Q.19 Answer:

Fluoroacetic acid $>$ Chloroacetic acid $>$ Bromoacetic acid

Solution:

The larger the electronegativity of halogen, the larger the acidity of substituted acetic acid.

Q.20 Answer:

The reduction of RCOOH with $\mathrm{LiAlH_4}$ produces $\mathrm{RCH_3}$.

Solution:

Explaining the options

(a) The product obtained is $\mathrm{RCH_2OH}$.

(c) This is due to the resonating effect amongst the two CO groups.

(d) The stabilization is due to the dispersal of negative charge of the carboxylate anion.

Q.21 Answer:

bicarbonate

Solution:

$\mathrm{CO_2}$ comes from $\mathrm{HCO_3^-}$

$$\mathrm{C_2H_5COOH + NaHCO_3 \longrightarrow C_2H_5COONa + H_2O + CO_2}$$

Q.22 Answer:

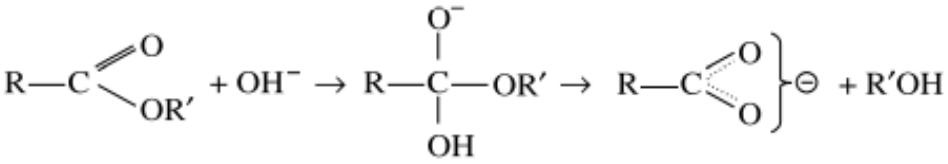
Alkaline hydrolysis of an ester follows second order kinetics and is thus $\mathrm{S_N2}$.

Solution:

Explaining the option

(a) The reaction is

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The reaction is irreversible because the resonance-stabilised carboxylate anion shows little tendency to react with an alcohol.

(b) The bond broken is the bond between oxygen and acyl carbon.

(d) An electron-withdrawing group stabilises the intermediate carboxylate anion, hence speeds up the rate of hydrolysis. So, the only correct option is (c)

Q.23 Answer:

The $\mathrm{p}K_{\mathrm{a1}}^{\circ}$ of a dicarboxylic acid is usually greater than the $\mathrm{p}K_{\mathrm{a}}^{\circ}$ of the comparable monoalkylcarboxylic acid.

Solution:

Explaining the option

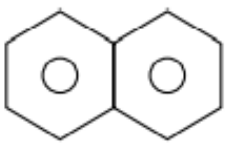
(a) The sp ^{2} hybridised carbon of $-\mathrm{C}=\mathrm{C}$ - attracts the bonded electrons more than the sp ^{3} hybridised carbon does. Consequently. $-\mathrm{C}=\mathrm{C}-$ is an acid-strengthening electron-withdrawing group.

(b) The $-\mathrm{COO}^{-}$ of the branched acid is shiclded from solvent molccules and cannot be stabilised as cffectively as can an unhindered anion.

(c) In trans isomer, the $-\mathrm{COO}^{-}$ is less shielded and is thus more stabilised by solvent.

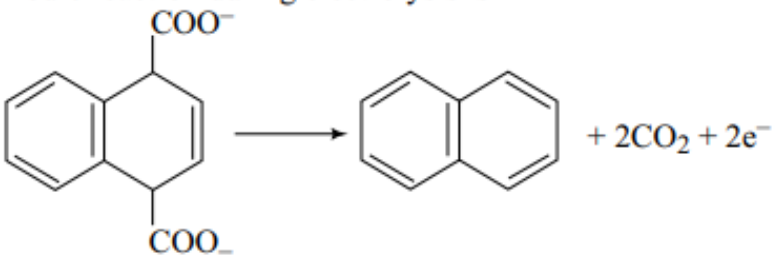
(d) The first ionization of a dicarboxylic acid is larger than that of monoalkylcarboxylic acid because of the electron-withdrawing COOH group of dicarboxylic acid. Hence, $\mathrm{p}K_{\mathrm{a1}}^{\circ}$ of dicarboxylic acid is smaller than $\mathrm{p}K_{\mathrm{a}}^{\circ}$ of monoalkylcarboxylic acid. So, the correct option is (d) since the statement is wrong

Q.24 Answer:



Solution:

Anodic reaction during electrolysis is



Q.25 Answer:

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$CH_3CH(Cl)COOH$

Solution:

Treatment with chlorine and a catalytic amount of phosphorus leads to the selective α-chlorination of carboxylic acids (HVZ reaction).

$$CH_3CH_2COOH + Cl_2 \xrightarrow{RedP} CH_3CHCl - COOH$$

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